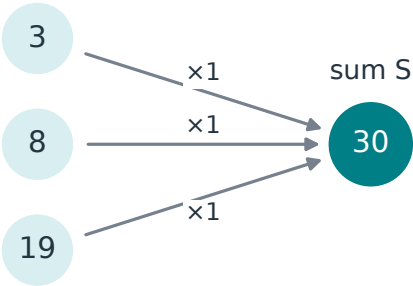
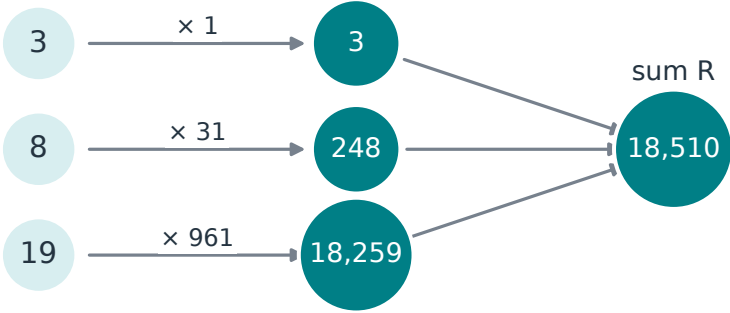


A. First query: give every entry weight 1



Choose $B = S + 1 = 31$.
Every hidden entry is smaller than B.

B. Second query: assign distinct digit weights



$3 + 248 + 18,259 = 18,510$
Each entry has its own base-31 position.

C. Divide by 31: pass the quotient onward and keep the remainder

